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June 9, 2025

The Honorable Mehmet Oz, MD
Administrator
Centers for Medicare & Medicaid Services
200 Independence Avenue, SW
Washington, DC 20201

RE: Request for Information: Health Technology Ecosystem [CMS-0042-NC]

Dear Administrator Oz,

The American Academy of Neurology (AAN) is the world's largest association of neurologists and neuroscience professionals, with over 40,000 members—and the leading voice on brain health. The AAN's mission is to enhance member career fulfillment and promote brain health for all. As you know, a neurologist is a doctor who specializes in the diagnosis, care, and treatment of brain, spinal cord, and nervous system diseases. These neurological diseases affect over one in three people worldwide and include Alzheimer's disease, stroke, concussion, epilepsy, Parkinson's disease, multiple sclerosis, peripheral neuropathy, migraine, and many more.

The AAN thanks the Centers for Medicare and Medicaid Services (CMS) for the opportunity to share feedback pursuant to the Request for Information about digital health products available to Medicare beneficiaries and the state of data interoperability and the broader health technology infrastructure. Much progress has been made to enhance secure and timely access to patient health information and the flow of data between clinicians and other health care stakeholders. Nevertheless, neurology providers experience ongoing challenges associated with interoperability, forcing them to spend significant time on administrative tasks instead of direct patient care.

As the quality and timeliness of health data available to patients through standards-based application program interfaces (API) increase, evolving digital health products will gain greater functionality and potential for enhancing the healthcare experience. The AAN believes the perspectives and needs of neurology providers and patients must be considered in any updates to regulatory frameworks pertaining to the health technology ecosystem.

PR-1: What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved digital health products for their patients?

a. *What are the current obstacles?*

Digital health products, including asynchronous telehealth technologies and wearables, are increasingly prevalent in neurological care. Certain devices and software programs allow neurology practitioners to care for patients outside the traditional clinic setting. Nevertheless, the AAN would like to make CMS aware of some of the multifaceted obstacles to appropriately leveraging digital health products.

First, the applicability and implementation of many wearables pertinent to neurology have not been widely studied. Among a series of products used for the management of stroke, migraine, depression, Alzheimer's

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Disease, and epilepsy, many studies did not demonstrate clinical effectiveness[1]. AAN members also report challenges with deciphering the options available to patients and ensuring sufficient time within each clinical visit to appropriately set up the device and counsel patients on their use. The market for consumer health products is presently so diverse and fragmented that patients with multiple comorbidities would need to use several devices (blood pressure monitor, smartwatch, connected weight scale, pulse oximeter, among others) and each of their associated smartphone applications to capture data of clinical benefit.[2] Third, the utility of certain digital health products is also limited without integration into the electronic health record (EHR). Fourth, there is high variability across healthcare systems with regards to internal standards and safety requirements. This could be improved by widespread standardization or third-party certification. Lastly, the AAN is concerned that the cost of integrating digital health products into care delivery and variable insurance coverage may reinforce health disparities among underserved patient populations.

b. *What information should providers share with patients when using digital products in the provision of their care?*

Conversations with patients about digital health products should emphasize the value, safety, reliability, and limitations of the product. The AAN believes that patients should understand the privacy implications of each product, including the data that is collected, how data will be used and stored, and who will have access to the data. Counseling patients on compliance and adherence to the product's use will also be necessary to avoid replicate testing. Further, the AAN believes that while the patient's provider is a valuable resource in educating patients about key considerations concerning usage of these products, providers should not be held liable for data access incidents stemming from the usage of these products or be required to specifically inform patients about data use or privacy implications for third-party direct to consumer devices.

The AAN believes that patients' understanding of the benefits and limitations of these products is of the utmost importance. Still, we remain concerned about the level of administrative burden that educating patients on these topics may present for neurology providers.

c. *What responsibilities do providers have when recommending the use of a digital product by a patient?*

In collaboration with health information technology (IT) staff and legal departments, healthcare providers should thoroughly vet digital health products for clinical effectiveness, safe usage, and cybersecurity before recommending products to patients. As with other types of therapy, health care providers should assess the indication, weigh risks and benefits, and explain potential disadvantages to their patients. With digital health products in particular, there is a need to evaluate the clinical appropriateness in alignment with the patient's needs, explain data handling, establish realistic expectations, and discuss ongoing monitoring and follow-up plans. The AAN believes that digital health tools should complement, but not entirely replace, the need for face-to-face neurological care.

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[1] Minen MT, Stieglitz EJ. Wearables for Neurologic Conditions: Considerations for Our Patients and Research Limitations. *Neurol Clin Pract*. 2021;11(4):e537-e543. doi:10.1212/CPJ.0000000000000971

[2] Phi NTT, Montori VM, Kunneman M, Ravaud P, Tran VT. Cumulative Burden of Digital Health Technologies for Patients With Multimorbidity: A Systematic Review. *JAMA Netw Open*. 2025;8(4):e257288. Published 2025 Apr 1. doi:10.1001/jamanetworkopen.2025.7288

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While the AAN believes there is promise in digital health products to advance neurological care, the landscape of these technologies is rapidly evolving, and integration will require evidence-based frameworks for use.

PR-2: What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

The AAN is committed to advocating for policies that promote patient-centered care while minimizing administrative burdens related to quality reporting, documentation, and billing processes. Annual updates to quality measurement reporting can be challenging to review, implement, and optimize within the clinical workflow. In addition, analytics and data reporting tools remain inaccessible to most physicians. The lack of interoperability and high costs of employing strategies to streamline the above activities present the primary barriers to implementation, especially for small neurology practices.
Artificial intelligence (AI) tools that generate progress notes, patient summaries, or assist with medical billing are promising, yet many are expensive and have not been programmed to aid clinicians who care for patients with rare neurological conditions.[3] Presently AI models that work well in primary care settings may perform differently for neurology providers and subspecialists. The AAN believes that AI should be used alongside clinicians for assistance, rather than replace medical judgement altogether. The AAN is also supportive of the Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology's (ASTP/ONC) previously stated goals to ensure that clinicians have the necessary information to determine whether a particular AI tool is fair, appropriate, valid, effective, and safe[4].

PR-3: How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

a. *Current challenges in accessing different data formats*

The AAN supports efforts to streamline EHR interoperability and promote access to crucial health data. The AAN has previously supported CMS's efforts to advance the use of standardized, Fast Healthcare Interoperability Resources (FHIR) based APIs and the adoption and implementation of the updated United States Core Data for Interoperability.

Although much progress has been made to harmonize EHR data in the last decade, challenges persist. Data shared via scanned documents or faxed records are difficult to incorporate into the EHR and are sometimes stored as PDFs in parallel databases. Often, data shared in this format are not electronically stored as discrete values, and the clinician must know how to search the EHR to find this information. Even with sophisticated health information exchange (HIE) tools, visit summaries with free-text notes originating from another office may not be fully viewable. Additionally, information (text, medications, orders, referrals, testing results) from inpatient consultations or visits are often transmitted as one large file that requires extra effort and time for the clinician to search for information of interest.

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[3] Schor NF. Equipping AI for Unbiased and Inclusive Neurology. *JAMA Neurol*. 2025;82(3):211-212. doi:10.1001/jamaneurol.2024.3954
[4] 89 FR 1192

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b. Impact on patient care and quality

It is crucial that patients' protected health information (PHI) is appropriately shared between providers, whether from a rural or community hospital to a tertiary care center or from a patient's primary care provider to a neurologist. Patient care is more cohesive when providers can easily reference medication lists, allergies, testing results, and office notes to understand what previous clinicians have assessed and determined. Health care quality and safety may suffer without appropriate health information exchange, leading to repeat testing and potential errors.

Health IT modules supporting public health data exchange

The AAN also supports efforts to enhance data accessibility across the care continuum, including for public health reporting and population health research. Doing so must not come at the expense of additional administrative burden on physicians or practice staff, and criteria for reporting should be discrete and automatically implementable within existing Certified EHR Technology (CEHRT). Given that public health reporting includes non-infectious conditions, such as multiple sclerosis, Parkinson's disease, and dementia, the data elements included in CEHRT must be appropriate for neurological diagnoses that may be required for reporting in various states. As standards continue to develop, the AAN agrees that establishing minimum functional capabilities and exchange standards for health IT to send and receive public health data is appropriate.

c. Technical barriers to full data accessibility

Full data accessibility is hindered by the wide variety of non-structured, non-standardized data formats, including scanned or faxed documents. The health information from these documents is often less accessible to health care providers because they are stored in different sections of the EHR and may not be incorporated into clinical workflows like other structured data, such as lab results. While full data access is necessary, the volume of data, including detailed imaging studies, video-EEG, or genome sequencing, can be daunting for many providers and is not always needed in every clinical encounter. Metadata and accessible, open-source, content summaries should be made available in addition to information on how and where to access the full dataset if needed.

d. Cost or privacy implications of making all data formats interoperable

AAN members report that the cost to transmit, store, and manage different data formats is increasing. The AAN is concerned that health care providers, especially those in private practice, will bear the burden of expenses related to digital health compliance. Although the AAN believes it is crucial for providers to obtain their patients' comprehensive health information in a timely fashion, we caution that the provider's liability should be limited because clinical decision support tools used for population management may vary by the end user. The AAN is concerned about potential impacts to clinical care should the availability, accuracy, or provenance of certain data, especially non-standardized or unstructured data, be compromised.

Additionally, the AAN is concerned with the impact of open API disclosure of highly sensitive patient data, such as genetic testing results. The AAN firmly believes that discretion is of the utmost importance in cases in which the results of high-risk genetic tests are conveyed to patients. In cases in which test results implicate a very high rate of

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developing a neurodegenerative disease for which there is no cure or effective treatment, such as Huntington's Disease, the AAN believes an exception to data sharing requirements is appropriate to ensure that patients and their families are not exposed to this information through an open API without proper counseling.

e. Priority level compared to other interoperability needs

The AAN has previously submitted comments in support of CMS-0057-P, which included several requirements to promote access to timely prior-authorization through implementing FHIR-based exchange of claims and encounter data, among other provisions. AAN members often cite burdens associated with prior authorization as a top administrative concern impacting patient care. Most physicians in the U.S. report that prior authorization requirements lead to care delays and in some instances, serious adverse events for patients under their care[5]. The AAN requests that CMS continue to improve electronic prior authorization processes and promote the simplification of data exchange between payers and providers.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

The AAN supports efforts to streamline administrative workflows that allow patients to access neurological care at the right place and time, and with a provider that best addresses their needs. Presently, auto-populating intake forms via a third-party product can be challenging because many pre-visit forms, even if they contain standard patient-reported outcome measures, are customized to the health care organization's EHR. The AAN is concerned that without rigorous testing for accuracy and reliability, third-party digital products could negatively impact patient data provenance and subsequently compromise patient care.

While understanding the utility of online scheduling tools, we are also concerned that enabling a third-party digital product may present cybersecurity risks to patient health information. This is especially true given that many EHR vendors already have fully integrated patient portals that allow patients to view and schedule appointments and complete pre-visit forms.

The AAN is unclear on the security framework for third-party APIs to prevent unauthorized disclosures after PHI is transmitted. The AAN understands that HIPAA-covered entities are not responsible for the security of PHI after it has been transferred to a third-party application at the direction of a patient[6], but requests clarification on the responsibilities of third-party applications to ensure that sensitive patient information remains protected.

Additionally, given the lack of a regulatory framework, the AAN is concerned about additional burdens on providers to ensure that patients are informed of potential data risks related to third-party products. While the AAN understands the importance of informed consent, patients may not reasonably understand the full implications of a disclosure of this sort. This would work counter to CMS's goal of promoting data exchange and could be potentially detrimental to the patient-provider relationship.

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[5] AMA Prior Authorization Physician Survey. American Medical Association. 2025. https://www.ama-assn.org/system/files/prior-authorization-survey.pdf
[6] 84 FR 7621

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PR-6: Is TEFCA currently helping to advance provider access to health information? Please provide specific examples.

The AAN remains supportive of the Trusted Exchange Framework and Common Agreement (TEFCA) as a nationwide mechanism to enhance the exchange of patient data. Recognizing that TEFCA has only recently been implemented, use cases for population health and specialty care have yet to be defined. Ensuring interoperability within neurological practices also remains relatively onerous. Many AAN members must register for and comply with various local, state, and specialty interoperability networks, often through vendors with a patchwork of functionality. To support universal interoperability within the United States, the AAN encourages CMS to consider how the agency can direct resources and incentives to accelerate specialty-specific use cases under TEFCA.

PR-7: What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize that could be reused for, or combined with, efforts needed to support interoperability?

As the digital health ecosystem evolves with an increasing volume of data available, the AAN would like to reiterate the challenges faced by many health care providers to process the influx of information as well as find the most pertinent data related to a patient's visit and condition(s). Accessing patient health data is not only constrained by the sheer volume of data available, but also the need for clinicians to sort through data from different modalities, including structured data, unstructured notes, scanned and faxed documents, diagnostic reports, and imaging studies. Further, healthcare providers must navigate liability considerations stemming from increased access to patient data.

AI tools that utilize natural language processing can mitigate the time clinicians spend sorting through patient health information by providing summaries and finding the most pertinent data points to support timely and appropriate medical decision making. At the individual patient level, challenges persist in finding health records for patients who seek care at different institutions. Simplifying patient identification across EHR systems will improve patient care coordination. This may be achieved by developing a nationwide patient identification strategy.

CMS could also consider encouraging incentives for digital health products and EHRs to use standardized and interoperable APIs, data dictionaries, and single sign-ons to reduce the need for multiple logins for each product and to promote easier access to all aspects of a patient's health information.

The AAN would like to note that some neurology practices prioritize data sharing differently, especially as it relates to the practice's financial and workforce capacity to do so. The AAN cautions CMS against policies that might create undue burdens without meaningfully impacting patient care. To promote data harmonization while also managing administrative burdens, the AAN encourages CMS to customize data sharing requirements that match the practice's size, specialty, and technical maturity. Leveraging automation and embedding data capture into clinical workflows, so that data sharing automatically occurs as a part of routine care, will help eliminate time-consuming manual processes allowing for more time to be dedicated to patient care. Lastly, the burden of increasing data

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availability may be mitigated in certain practices by encouraging the creation of dedicated physician informaticists who are passionate about digital health and interoperability.

PR-8: What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

a. *What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?*

Implementing Bulk FHIR data exports can streamline clinical quality data submissions, but the data elements must be precisely defined and relatively standardized across different EHRs. Presently, AAN members report challenges reconciling PHI between different organizations despite ongoing efforts to improve the mapping process.

Before CMS can perform quality metrics calculations leveraging Bulk FHIR data exports, each quality data element must be defined and standardized, which includes harmonizing codes and units for structured and unstructured data. Tools to reliably pull data from disparate EHR systems must also be validated.

b. *In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?*

The above challenges related to data standardization must first be resolved to dually-purpose the interoperability and quality reporting responsibilities. Secondly, CMS could consider aligning measures with data that are already captured during routine patient care. For example, many metrics such as emergency department wait times are not part of standard clinical notes and require separate workflows to report. In contrast, tracking timestamps for shunt placement and documenting related infection events would leverage data that clinicians already monitor. Implementing quality measures that can be automatically extracted from existing structured and unstructured documentation allows interoperability investments to fulfill reporting needs without extra provider burden.

PR-12: Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

Although supportive of the rationale for the Preventing Harm Exception, the AAN has previously expressed concern regarding the impact of open API disclosure of highly sensitive patient data. While this exception should certainly remain in place, the AAN strongly believes that discretion is of the utmost importance in cases where high-risk laboratory or imaging are being performed. The ASTP/ONC's 21st Century Cures Act final rule and subsequent regulations that build upon information blocking allow for a Preventing Harm Exception, whereby "the actor engaging in the practice must hold a reasonable belief that [information blocking] will substantially reduce a risk of harm to a patient or another natural person that would otherwise arise from the access, exchange, or use of electronic health information affected by the practice." Health care providers are permitted, on a case-by-case basis, to meet the Preventing Harm Exception if the rationale for delaying the release of the results is documented for the individual patient. Situations such as genetic testing that may implicate a chance of developing a neurodegenerative disease for which there is no cure or effective treatment require appropriate counseling and

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context regardless of the outcome of the testing. In fact, patient surveys demonstrate that in situations of potentially debilitating or terminal illnesses, more than half of patients prefer to speak with their health care provider before reviewing the results in a patient portal[7].

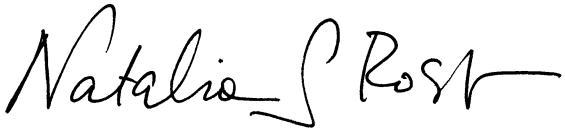
Lastly, the Preventing Harm Exception leaves a gray area in the rapidly growing consumer market of self-service testing. The AAN recognizes that in both the clinical and consumer space, tests such as whole-body MRI scans or genome sequencings may leave individuals unprepared for results that implicate the presence of biomarkers of potential neurodegenerative disease.

The AAN is concerned that relying on "reasonable belief" and "professional judgment" standards does not sufficiently address the significant psychological and emotional harm facing patients who receive results absent appropriate counseling. The AAN recommends that this exception be expanded to permit the formation of organization-wide policies to delay the release of results for terminal or other highly sensitive testing until the provider has discussed the results and logged the discussion with the patient in the medical record. The AAN would like to caution against the formation of any policies that may restrict the sharing of health information between healthcare organizations that care for mutual patients.

Conclusion

The AAN greatly appreciates the opportunity to provide CMS with input on the current state of data interoperability and the broader health technology infrastructure. Please contact Matt Kerschner, the AAN's Director of Regulatory Affairs and Policy, at mkerschner@aan.com and Michaela Read, the AAN's Practice and Healthcare Technology Program Manager, at mread@aan.com with any questions or requests for additional information.

Sincerely,



Natalia S. Rost, MD, MPH, FAAN, FAHA
President, American Academy of Neurology

[7] Preventing Patient Harm. American Medical Association. 2022. https://www.ama-assn.org/system/files/patient-privacy-survey-results-preventing-patient-harm.pdf